

UE23CS320B - Capstone Project Phase - 2

Project Progress Review #2

Project Title : AI Phishing Simulation via Hybrid Agent-Based Modeling
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Outline

- Abstract and scope.
- Suggestions from phase 2 review 1
- Design Approach.
- Design constraints, Assumptions and Dependencies
- Proposed methodology/Approach
- Architecture
- Design Description
- Technology used
- Dataset Exploration (if any)/Tool Exploration(as Applicable)
- Data Pre-processing
- References
- Any other Information

Abstract and Scope

- Introduces a Hybrid Agent-Based Simulation approach to safely study AI-enhanced phishing
- Deploy these datasets inside a simulated workplace environment.
- Develop AI agents that simulate human employee behavior.
- Configure agents using real-world human factors such as Fatigue, stress, Cognitive load
- Use email labels and behavioral parameters to model probability of clicking phishing emails under different simulated human conditions.
- Evaluate detection evasion strategies by measuring how long phishing campaigns remain undetected.
- Compare effectiveness of LLM phishing vs Human phishing in realistic behavioral scenarios, and use these findings to support cybersecurity training, detection research and risk analysis.

Suggestions from Phase2 review 1

- To get started with the agent
- To gain insights from social science/psychology papers pertaining to the factors and effects of worker performance under stress, fatigue and other factors, which will help us understand and build our agent.

Design Approach

Hybrid Agent Based Simulation with Behavioral Agent modelling:

- Use an agent based simulation environment to replicate workplace interactions
- AI agents now use LLMs for processing, and will be built to simulate employees with configurable behavioral parameters.
- Agents are influenced by real-world factors such as fatigue, stress, and cognitive load. The primary approach is to pair the LLM processing with python implementation that deals with agent state and perception. Another approach that we looked into was to let the LLM handle both the cue processing and agent states.
- Mails (Phishing + Benign) will be injected into the environment, and the agent processes these mails with cues extracted and factors in it's behavioral state like stress, energy, performance, perception level, etc.

Why this approach?

- Phishing success depends heavily on human behaviour, not just email content
- Real-world testing of phishing is unethical and risky; Simulation allows safe experimentation at scale
- Agent-based models allow testing multiple human conditions simultaneously
- Helps evaluate phishing effectiveness in realistic workplace scenarios

Design Constraints, Assumptions & Dependencies

- The LLM being used has a limit for API requests, so full cue extraction process will be done in multiple sessions. On exhaustion of limits, system falls back into a regex based process that will be less accurate.
- Agent traits (fatigue, job performance, etc.) are assumed and derived from literature defined behavior and ranges.
- Some assumptions made in the agent implementation:
 - LLM cue extraction is treated as ground truth
 - Workload and time pressure increase linearly across the workday (8am to 4pm)
 - A suspicion threshold of 2-6 cues is a reasonable proxy for human skepticism variance
 - Note: We can and will improve upon these implemented assumptions in further reading and development

Design Constraints, Assumptions & Dependencies

- Dependencies:
 - Gemini API (gemini-2.5-flash): Cue extraction from emails, falls back to regex if unavailable
 - data/cue_cache/ — To avoid re-calling API on re-runs
 - google-genai, pandas, numpy — Core runtime

Proposed Methodology / Approach

- We orchestrate a triple agent loop across three dimensions (agents, workday hours, email samples) that produces 37,500 data points
- The LLM will first extract cues from these mails (9 cues for now, we will be adding more cues as we go). On limit exhaustion, the system falls back to regex approach.
- The stochastic decision loop takes place at each data point, using which cues are read based on randomized flawed perception level and factors like job performance, fatigue, etc.

Proposed Methodology / Approach

- Benefits
 - Easily scale the system to more mails, more agents
 - Easy to analyse quantify click through rate for each time increment
 - The use of seeded random generation ensures that complex behavioral simulations remain deterministic and auditable.
- Drawbacks
 - API quota can limit the volume of rich extraction and may have to use regex fallback at times that can be less accurate
 - Over-reliance on simple threshold can lead to mails being mis-labelled

Architecture

- Provide high-level design view of the system.
- Identify the logical user groups, application components, data components, and interfacing systems. Illustrate the collaboration and interaction between the major components.

Design Description

Add as many slides as required to cover the following aspects(whichever is applicable):

1. Master class diagram
2. ER Diagram
3. User Interface Diagrams/ Use Case Diagrams
4. Report Layouts
5. External Interfaces

Technologies Used

What technologies you plan to use and why

Data Exploration

Dataset details:

- Source and link
- Size and Type of data
- Time Period(if applicable)

Tool Exploration

Tools and framework as applicable to your project

Data Preprocessing

Plan and progress on data preprocessing

References

Provide references pertaining to your research according to IEEE format.

Example:

[1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955. (*references*)

Any other information

Provide any other information you wish to add on.

Note: Changes can be made in the template, with the consent of the guide for inclusion of any other information.

Thank You